

order Fundamentals of Astrodynamics 2nd ed.

Comments: we look at the motion of bodies in space



Focus on translational motion (particles)

↘ attitude dynamics is AE 6356

Almost everything we do has roots in the N-body problem

N-Body problem

↳ described by Newton in 1687

↳ much of what we need centers around a few key results:

↳ 1. Newton's Second Law: Time rate of change of linear momentum is proportional to the force applied

$$\dot{\vec{p}} = \frac{d}{dt}(m\vec{v}) = \underbrace{\dot{m}}_0 \vec{v} + \underbrace{m\dot{\vec{v}}}_{m\vec{a}} \rightarrow \dot{\vec{p}} = m\vec{a}$$

when $\dot{m}=0$

↓
 $\vec{F} = \dot{\vec{p}} = m\vec{a}$

* Note: This law is only valid in an inertial frame



Frame that is not rotating or accelerating



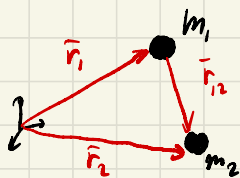
Brush up on kinematics and
Reynolds transport

A Frame in which forces on particles
cause no acceleration

2. Newtons Law of Universal Gravitation:

The gravitational attraction force between two
objects is proportional to their masses and
inversely to the distance between them.

$$F = G \frac{m_1 m_2}{r^2} \quad G = 6.674 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$



$$\bar{r}_{12} = \bar{r}_2 - \bar{r}_1$$

$\bar{F}_{12}$ at m_1 due to m_2

$$\bar{F}_{12} = G \frac{m_1 m_2}{r_{12}^2} \frac{\bar{r}_{12}}{r_{12}} \quad r_{12} = \|\bar{r}_{12}\|$$

$$\bar{F}_{12} = G \frac{m_1 m_2}{r_{12}^3} \bar{r}_{12}$$

$$\text{likewise: } \bar{F}_{21} = G \frac{m_1 m_2}{r_{21}^3} \bar{r}_{21} \Rightarrow \bar{F}_{12} + \bar{F}_{21} = 0$$

From this we obtain a pair of equations:

$$m_1: \quad \bar{F}_{12} = G \frac{m_1 m_2}{r_{12}^3} \bar{r}_{12} = G \frac{m_1 m_2}{\|\bar{r}_2 - \bar{r}_1\|^3} (\bar{r}_2 - \bar{r}_1) = m_1 \ddot{\bar{r}}_1$$

$$m_2: \quad \bar{F}_{21} = G \frac{m_1 m_2}{\|\bar{r}_1 - \bar{r}_2\|^3} (\bar{r}_1 - \bar{r}_2) = m_2 \ddot{\bar{r}}_2$$

6 Second order
Differential equations

$\ddot{x} + x - 3 = 0$ Second order

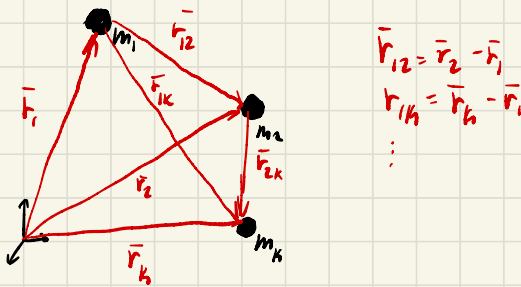
$\dot{x} + x - 3 = 0$ First order

12 degrees of freedom so we need 12 constants of integration



Generally: $6N$ DOFs for N -bodies

Let's generalize these laws for the N body problem:



For the i^{th} body of the system:

$$\vec{F}_i = G \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^2} \vec{r}_{ij} \quad , \quad \text{where} \quad \sum_{i=1}^N \vec{F}_i = 0$$

Is there enough info for a closed form solution for $n \geq 2$?

Let's assume all particles have a constant mass $\dot{m}_i = 0$

We can apply Newton's second law:

$$\sum \vec{F}_i = \sum m_i \ddot{\vec{r}}_i = 0$$

$$\downarrow \text{since } \frac{d}{dt} m = 0$$

$$\downarrow \frac{d^2}{dt^2} (\sum m_i \vec{r}_i) = 0$$

Let's define the center of mass as $\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$ and $\sum m_i \neq 0$

$$\frac{d^2}{dt^2} (\sum m_i \bar{r}_{cm}) = \sum m_i \frac{d^2}{dt^2} \bar{r}_{cm} = 0$$

$\uparrow$ not 0 $\uparrow$ must be zero

$\ddot{\bar{r}} = 0 \rightarrow$ system CM cannot accelerate

$\dot{\bar{r}}_{cm} = \bar{c}_1 \rightarrow$ linear momentum of the system is conserved

$\bar{r}_{cm} = \bar{c}_1 t + \bar{c}_2 \rightarrow$ System CM can move with constant velocity

↓

Knowing $\bar{r}_{cm}(t_0)$ and $\dot{\bar{r}}_{cm}(t_0)$: you know 6 constants of integration ($\bar{c}_1$ and $\bar{c}_2$) associated with conservation to the systems linear momentum.

Also consider angular momentum: look at torque...

$\sum \bar{r}_i \times \bar{F}_i$ consider $n=2$

$$\begin{aligned}
 \bar{r}_1 \times \bar{F}_1 + \bar{r}_2 \times \bar{F}_2 &= \bar{r}_1 \times \frac{Gm_1 m_2}{r_{12}^3} (\bar{r}_2 - \bar{r}_1) + \bar{r}_2 \times \frac{Gm_1 m_2}{r_{21}^3} (\bar{r}_1 - \bar{r}_2) \\
 &= \frac{Gm_1 m_2}{r_{12}^3} \left(\bar{r}_1 \times \bar{r}_2 - \cancel{\bar{r}_1 \times \bar{r}_1}^0 + \bar{r}_2 \times \bar{r}_1 - \cancel{\bar{r}_2 \times \bar{r}_2}^0 \right)
 \end{aligned}$$

$\swarrow \quad \searrow$
 $(-)$

$$= \bar{0} \quad \text{In General... } \sum \bar{r}_i \times \bar{F}_i = \bar{0}$$